Exponential Functions & Growth Models

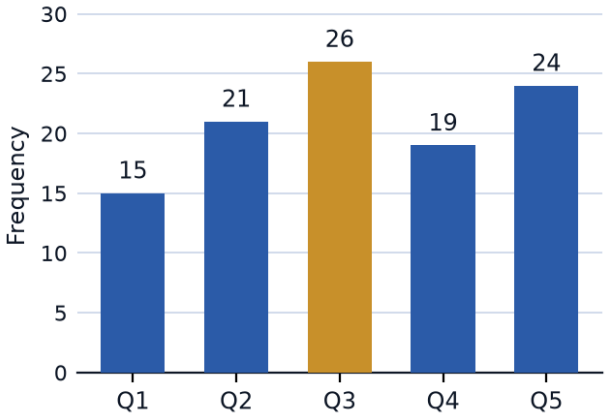
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponential growth and decay basics**

Classify each function as exponential growth or decay, and state the growth/decay factor: $f(x) = 5(1.2)^x$ and $g(x) = 5(0.8)^x$.

2 **[Foundational]** • **Target: Exponential growth and decay basics**

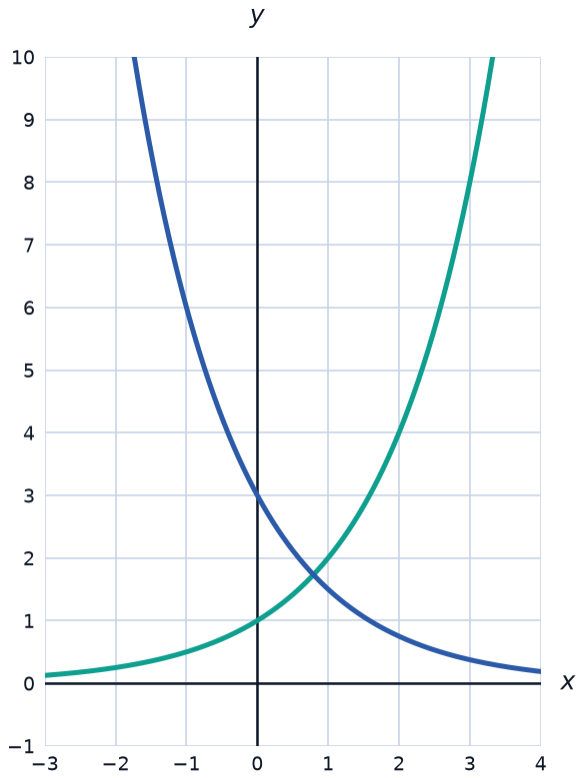
A population of bacteria is modeled by $P(t) = 1000(2)^{t/3}$, where t is in hours. Interpret the meaning of the exponent's structure, and find the population after 9 hours.



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3 [Foundational] • Target: Visualizing exponential growth

The graph shows $f(x) = 2^x$ and $g(x) = 3(0.5)^x$, one growing and one decaying.



a Identify which curve represents growth and which represents decay.

b State the horizontal asymptote shared by both curves.

4 [Foundational] • Target: Finding an exponential model from data

An investment grows from 2000 dollars to 2420 dollars in 2 years under exponential growth $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

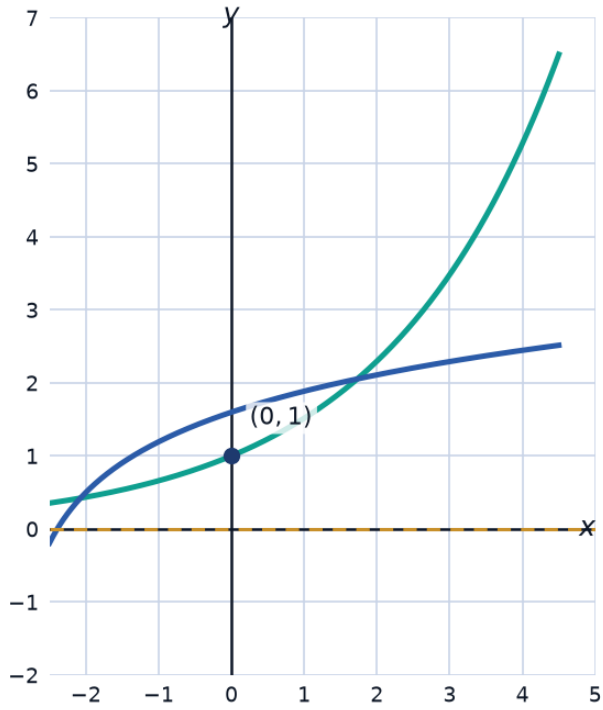
5 [Foundational] • Target: Finding an exponential model from data

A radioactive sample decays from 80 g to 45 g over 6 years. Find the annual decay factor b in the model $A(t) = 80\,b^t$, to four decimal places.

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6 [Foundational] • Target: Continuous growth and the number e

A quantity grows continuously according to $A(t) = 500e^{0.03t}$. Find the amount after 20 years, to the nearest whole number.



7 [Foundational] • Target: Continuous growth and the number e

Explain the difference between the discrete growth model $A(t) = A_0(1 + r)^t$ and the continuous growth model $A(t) = A_0e^{kt}$, and state how r and k are related when both models describe the same annual growth rate r.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing growth rates: exponential vs. polynomial

Explain why, for large enough x, any exponential growth function $f(x) = b^x$ (with $b > 1$) will eventually exceed any polynomial function $g(x) = x^n$, no matter how large the exponent n is. Illustrate with $f(x) = 1.01^x$ vs. $g(x) = x^{10}$.

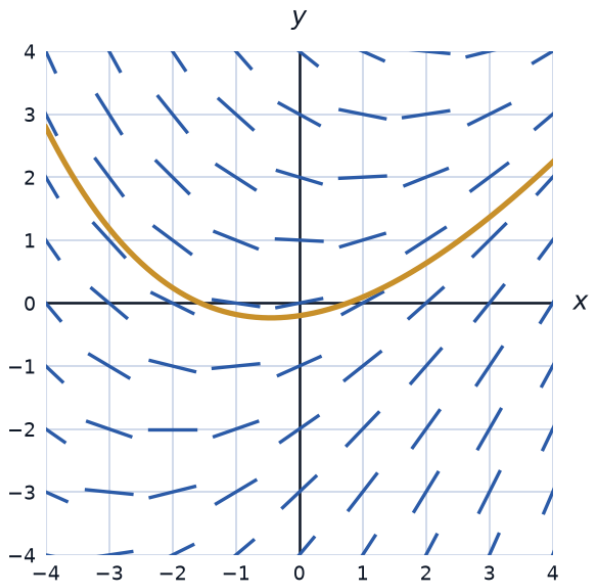
9 [Intermediate] • Target: Applications: half-life and doubling time

A substance has a half-life of 8 days. Write its decay model $A(t) = A_0\left(\frac{1}{2}\right)^{t/8}$, then find what fraction of the original amount remains after 20 days.

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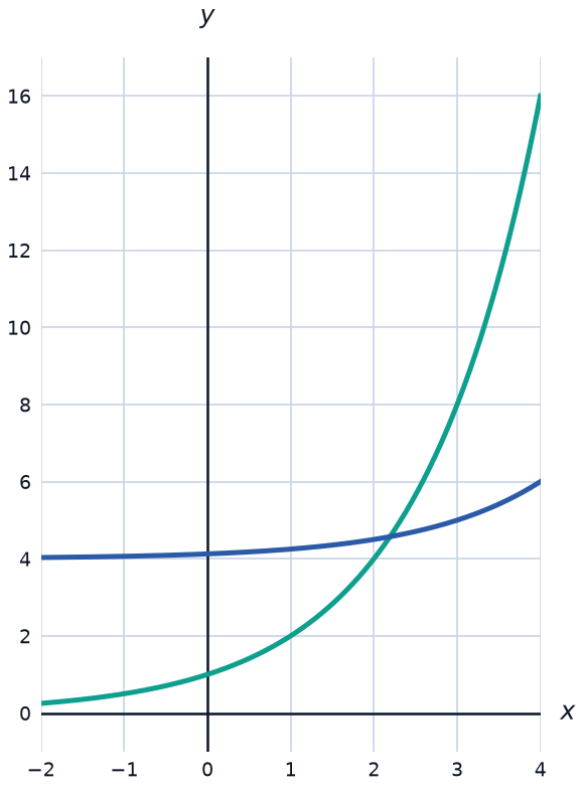
10 [Intermediate] • Target: Applications: half-life and doubling time

A city's population doubles every 15 years and currently stands at 40{,}000. Write an exponential model $P(t)$ and predict the population after 45 years.



11 [Intermediate] • Target: Transformations of exponential functions

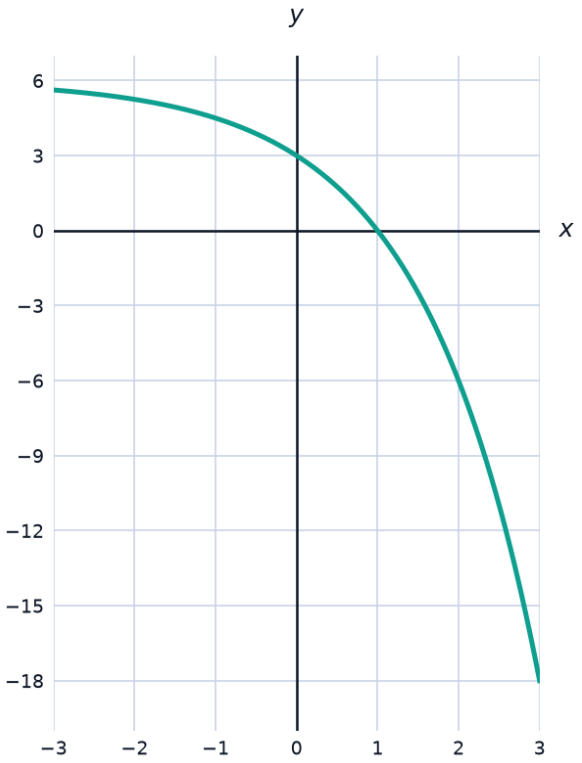
Describe how the graph of $g(x) = 2^{x-3} + 4$ is transformed from the parent function $f(x) = 2^x$, and state the new horizontal asymptote.



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12 [Intermediate] • Target: Transformations of exponential functions

Find the horizontal asymptote and y-intercept of $h(x) = -3(2)^x + 6$.



13 [Intermediate] • Target: Compound interest and financial models

A savings account earns 4% annual interest compounded quarterly, modeled by $A(t) = A_0\left(1 + \frac{0.04}{4}\right)^{4t}$. If 5000 dollars is deposited, find the balance after 3 years, to the nearest dollar.

14 [Intermediate] • Target: Compound interest and financial models

Explain, without computing exact values, why compounding more frequently (e.g. daily instead of quarterly) at the same nominal annual rate produces a slightly larger balance after the same number of years.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

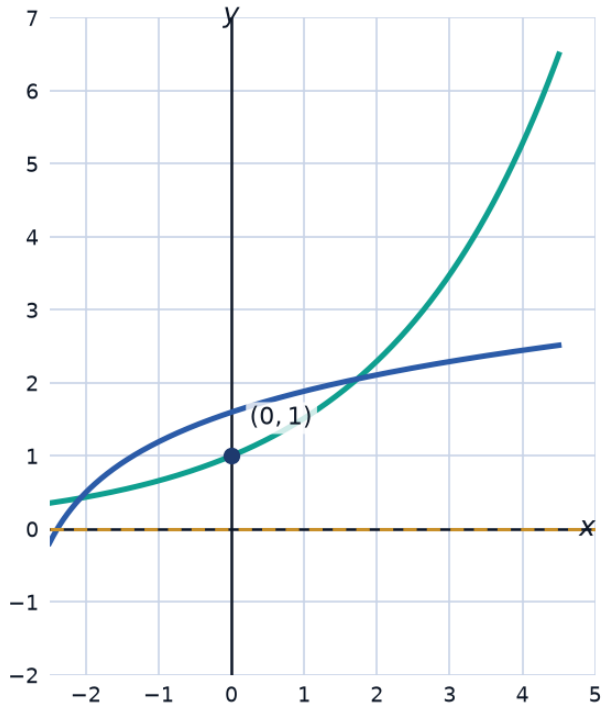
15 [Challenging] • Target: Finding an exponential model from data • Cross-Lesson: 4.03 Parametric Equations & Modeling

The half-life of Carbon-14 is 5730 years. (a) Write $A(t)$. (b) How much of 200 g remains after 11 460 years? (c) When is only 10% left?

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16 [Challenging] • Target: Continuous growth and the number e • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A cup of coffee at 92°C is placed in a room kept at a constant 20°C. After 10 minutes, the coffee has cooled to 68°C. Using Newton's Law of Cooling $T(t) = 20 + 72b^{t/10}$, find the exact temperature after 20 minutes and when the coffee reaches 36°C.



17 [Challenging] • Target: Comparing growth rates: exponential vs. polynomial • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Given that $\cos\alpha = -\frac{5}{13}$ with $\frac{\pi}{2} < \alpha < \pi$, find the exact values of $\sin(2\alpha)$, $\cos(2\alpha)$, and $\tan(2\alpha)$.

18 [Challenging] • Target: Applications: half-life and doubling time • Cross-Lesson: 1.03 Rates of Change & Concavity

Evaluate the exact value of $\sin\left(\arccos\left(-\frac{3}{5}\right)\right)$ and $\tan\left(\arcsin\left(\frac{8}{17}\right)\right)$.

19 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Analyze the polar curve $r(\theta) = 2 - 4\cos\theta$ for $0 \le \theta < 2\pi$. Identify the curve type, find the maximum distance from the pole, and determine the exact angles θ where the curve passes through the pole.

20 [Challenging] • Target: Compound interest and financial models • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Solve the trigonometric equation $2\cos^2 x - \sin x - 1 = 0$ for all x in the interval $[0, 2\pi)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exponential growth and decay basics

A population is modeled by $P(t) = 900(3)^{t/3}$, where t is in years. Find $P(6)$.

a. 2700

b. 5400

c. 8100

d. 24300

22 [Foundational] • Target: Visualizing exponential growth

A population is modeled by $P(t) = 700(3)^{t/3}$, where t is in years. Find $P(3)$.

a. 2800

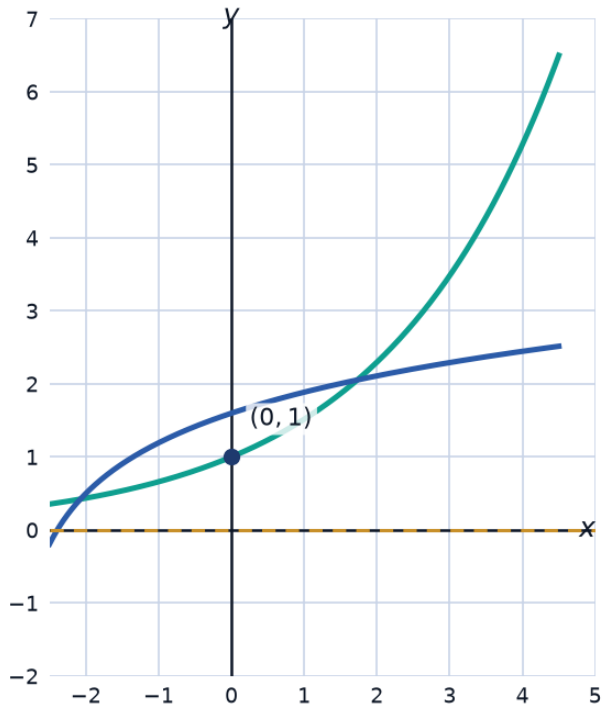
b. 6300

c. 1400

d. 2100

23 [Foundational] • Target: Finding an exponential model from data

A population is modeled by $P(t) = 700(2)^{t/2}$, where t is in years. Find $P(4)$.



a. 5600

b. 1400

c. 2100

d. 2800

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24 [Foundational] • Target: Continuous growth and the number e

A population is modeled by $P(t) = 500(3)^{t/3}$, where t is in years. Find $P(3)$.

a. 1000

b. 2500

c. 4500

d. 1500

25 [Foundational] • Target: Comparing growth rates: exponential vs. polynomial

A population is modeled by $P(t) = 900(3)^{t/5}$, where t is in years. Find $P(15)$.

a. 26100

b. 8100

c. 24300

d. 72900

26 [Foundational] • Target: Applications: half-life and doubling time

A population is modeled by $P(t) = 200(2)^{t/5}$, where t is in years. Find $P(20)$.

a. 3200

b. 6400

c. 1600

d. 3600

27 [Foundational] • Target: Transformations of exponential functions

A population is modeled by $P(t) = 500(2)^{t/5}$, where t is in years. Find $P(20)$.

a. 8500

b. 8000

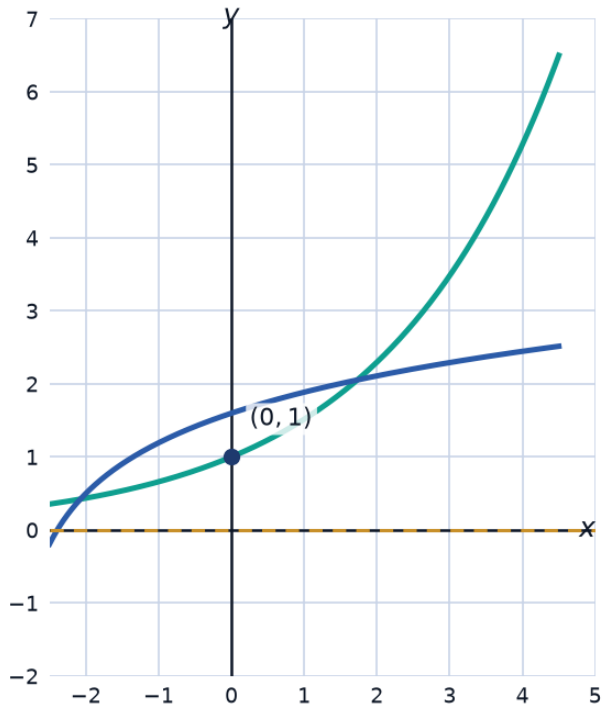
c. 4000

d. 16000

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28 [Foundational] • Target: Compound interest and financial models

A substance has a half-life of 6 days. What fraction of the original amount remains after 18 days?



a. $\frac{1}{4}$

b. $\frac{1}{16}$

c. $\frac{3}{8}$

d. $\frac{1}{8}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Exponential Growth and Decay Fundamentals

A substance has a half-life of 5 days. What fraction of the original amount remains after 15 days?

a. $\frac{1}{16}$

b. $\frac{1}{4}$

c. $\frac{1}{8}$

d. $\frac{3}{8}$

30 [Intermediate] • Target: Doubling Time & Half-Life Calculations

A substance has a half-life of 6 days. What fraction of the original amount remains after 12 days?

a. $\frac{5}{16}$

b. $\frac{1}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

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31 [Intermediate] • Target: Equivalent Exponential Forms & Conversions

A substance has a half-life of 7 days. What fraction of the original amount remains after 21 days?

a. $\frac{1}{8}$

b. $\frac{3}{8}$

c. $\frac{1}{16}$

d. $\frac{1}{4}$

32 [Intermediate] • Target: Comparing Linear and Exponential Growth

A substance has a half-life of 4 days. What fraction of the original amount remains after 4 days?

a. $\frac{5}{8}$

b. 1

c. $\frac{1}{4}$

d. $\frac{1}{2}$

33 [Intermediate] • Target: Exponential growth and decay basics

A substance has a half-life of 5 days. What fraction of the original amount remains after 25 days?

a. $\frac{5}{32}$

b. $\frac{1}{64}$

c. $\frac{1}{16}$

d. $\frac{1}{32}$

34 [Intermediate] • Target: Visualizing exponential growth

A substance has a half-life of 9 days. What fraction of the original amount remains after 45 days?

a. $\frac{1}{64}$

b. $\frac{1}{16}$

c. $\frac{1}{32}$

d. $\frac{5}{32}$

35 [Intermediate] • Target: Finding an exponential model from data

An investment grows from 200 dollars to 128 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

a. $-\frac{1}{10}$

b. $-\frac{1}{5}$

c. $\frac{1}{5}$

d. $\frac{4}{5}$

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36 [Intermediate] • Target: Continuous growth and the number e

An investment grows from 200 dollars to 162 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

a. $\frac{9}{10}$

b. $\frac{1}{10}$

c. 0

d. $-\frac{1}{10}$

37 [Intermediate] • Target: Comparing growth rates: exponential vs. polynomial

Find the horizontal asymptote and y-intercept of $h(x) = 5(2)^x + 1$.

a. Horizontal asymptote $y = 6$; y-intercept (1).

b. Horizontal asymptote $y = 1$; y-intercept (6).

c. Horizontal asymptote $y = 1$; y-intercept (5).

d. Horizontal asymptote $y = 0$; y-intercept (6).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications: half-life and doubling time • Cross-Lesson: 1.03 Rates of Change & Concavity

Find the horizontal asymptote and y-intercept of $h(x) = 3(2)^x - 4$.

a. Horizontal asymptote $y = -1$; y-intercept (-4).

b. Horizontal asymptote $y = -4$; y-intercept (-1).

c. Horizontal asymptote $y = -4$; y-intercept (3).

d. Horizontal asymptote $y = 0$; y-intercept (-1).

39 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 9$. If 5000 dollars is deposited, find the balance to the nearest dollar.

a. 7167

b. 7117

c. 5000

d. 7067

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40 [Challenging] • Target: Compound interest and financial models • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 13$. If 1000 dollars is deposited, find the balance to the nearest dollar.

a. 1000

b. 1515

c. 1665

d. 1765

41 [Challenging] • Target: Exponential Growth and Decay Fundamentals • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Which function represents exponential decay with an initial value of 450 and a decay rate of 8% per year?

a. $f(t) = 450(0.92)^t$

b. $f(t) = 450(1.08)^t$

c. $f(t) = 450(0.08)^t$

d. $f(t) = 450 - 0.08t$

42 [Challenging] • Target: Exponential Growth and Decay Fundamentals • Cross-Lesson: 4.01 Inverse & Composite Functions

If a quantity grows continuously at an annual rate of 5%, what is the continuous growth model for an initial investment of 1200 dollars?

a. $A(t) = 1200e^{0.05t}$

b. $A(t) = 1200(1.05)^t$

c. $A(t) = 1200(0.05)^t$

d. $A(t) = 1200e^{1.05t}$

43 [Challenging] • Target: Doubling Time & Half-Life Calculations • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

If the doubling time of an investment is 12 years, what is the annual growth factor b in $A(t) = A_0b^t$?

a. $2^{1/12}$

b. $12^{1/2}$

c. 2×12

d. $\frac{2}{12}$

44 [Challenging] • Target: Equivalent Exponential Forms & Conversions • Cross-Lesson: 4.03 Parametric Equations & Modeling

If $P(t) = 250e^{0.08t}$, what is the equivalent discrete annual growth factor rounded to four decimal places?

a. 1.0833

b. 1.0800

c. 1.0408

d. 1.1664

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45 [Challenging] • Target: Comparing Linear and Exponential Growth • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A linear function $L(x) = 50x + 100$ and an exponential function $E(x) = 2^x$ are compared for large $x > 0$. Which statement is true?

a. Eventually $E(x) > L(x)$ for all sufficiently large x

b. Eventually $L(x) > E(x)$ because its initial rate of change is 50

c. They will intersect infinitely many times

d. $L(x)$ and $E(x)$ grow at the same rate asymptotically

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